Crowded Out: Measuring Rental Market Pressure with Household Density

[1,2*]

[1*], , , , , , .

Corresponding author(s). E-mail(s): ;

Abstract

We introduce the Rental Density Index (RDI), a household-level measure of rental market tightness defined as the number of occupants per renter household. We show that deviations of renter household density from demographic and unit-size fundamentals— excess crowding—contain predictive information about subsequent rent growth across U.S. metropolitan areas. Interpreted within a stock-flow adjustment framework, excess crowding functions as a slow-moving state variable that summarizes accumulated imbalance between household formation and housing supply. Using panel data for the 100 largest U.S. metropolitan areas, we find that markets with higher excess crowding experience stronger subsequent real rent adjustment, even after conditioning on new supply and standard demand indicators. The predictive content of excess crowding is strongest in dense, space-constrained markets and is absent in placebo tests, reinforcing its interpretation as latent demand pressure rather than a mechanical correlation. strongest in dense, space-constrained markets and disappears under placebo tests, reinforcing its interpretation as latent demand pressure rather than a mechanical correlation. Our results highlight household consolidation as an important non-price margin of adjustment in rental markets and demonstrate that renter household density provides information about market tightness that is not captured by prices, vacancies, or absorption alone.

Keywords: density, latent demand, multifamily housing, forecasting, rent growth

1 Introduction

Housing availability and affordability has become a central policy concern in many U.S. metropolitan areas. Over several decades, housing production has grown slower than population, contributing to a national housing shortfall. While the median estimate of this shortfall is 3.9 million units, recent estimates range widely–from 1.5 million to over 20 million units–reflecting differences in methodology and underlying assumptions (Table A). Consistent with these pressures, a substantial share of renter households devote a large fraction of income to housing: recent Census estimates indicate that nearly half of renter households spend more than 30 percent of income on rent, a commonly used threshold for defining housing-cost burden U.S. Census Bureau (2022). Over the same period, shelter costs rose faster than overall consumer prices; between 2000 and 2024, the CPI for shelter increased by 107 percent, compared with a 71 percent increase for non-shelter items Federal Reserve Bank of St. Louis (2024). Despite these national trends, several high-growth metropolitan areas have recently experienced rent declines following periods of rapid new construction Mott (2024). Together, these patterns indicate that rent outcomes reflect both long-run supply constraints and short-run market adjustments that vary across locations.

Measures commonly used to characterize demand in multifamily housing markets, such as occupancy rates and net absorption, are informative about near-term conditions but are closely tied to the existing housing stock. Occupancy rates are mechanically capped, and net absorption primarily reflects the pace at which newly delivered units are leased rather than the intensity of underlying household formation pressure, particularly when supply adjusts slowly Mueller and Laposa (1999). As a result, these indicators provide limited insight into how demand is expressed in markets where households respond along other margins, such as by altering living arrangements Gabriel and Nothaft (2001); Sirmans et al. (1991); Pyhrr et al. (1999). This limitation complicates efforts to distinguish whether observed rent movements primarily reflect supply expansion, changes in household formation behavior, or shifts in the desirability of particular locations Pennington (2021); Molloy (2022). The distinction, however, is central to ongoing debates over the relative roles of supply constraints Saiz (2010) and demand for high-amenity locations Davidoff (2015).

This paper studies rental markets through the lens of a stock-flow adjustment process in which housing supply evolves slowly, while population and household formation respond more quickly to economic conditions. Within this framework, we focus on renter household density (called the Rental Density Index, or, RDI), defined as the average number of occupants per renter household in a metropolitan area. Changes in household density reflect how renters adjust their consumption of housing space in response to changes in prices or availability. When housing becomes scarce or expensive, households may delay forming independent households, they may add roommates, or they may consolidate living arrangements. When conditions ease, households tend to de-densify through expansion. These adjustments can occur even when measured occupancy remains high, making the RDI a potentially informative indicator of space-market pressure.

Variation in renter household density reflects both long-run structural factors and short-run market imbalances. Age composition, family structure, and unit size shape baseline household sizes across metropolitan areas, while departures from these fundamentals arise when supply does not adjust quickly to shifts in demand. To separate these components, we decompose renter household density into a predictable component driven by demographic structure and unit size, and a residual component, excess crowding, that captures density beyond what local fundamentals would imply. We interpret excess crowding as a state variable summarizing latent demand pressure within a stock-flow adjustment framework. In this paper, we use the term "latent demand" in a narrow sense: not as unmet preferences at prevailing prices, but as accumulated pressure arising when population and household formation adjust more quickly than housing supply. Excess crowding reflects how this pressure is temporarily absorbed through household consolidation rather than immediate price adjustment.

This perspective is motivated by broader changes in population density, rental costs, and tenure choice over recent decades. Between 2000 and 2022, population density in high-income countries increased by approximately 12 percent, while density in the United States increased by roughly 19 percent The World Bank (2025). Over the same period, rents rose faster than both wages and non-shelter prices Feiveson et al. (2024); Federal Reserve Bank of St. Louis (2024), and younger cohorts increasingly delayed homeownership and remained in the rental sector longer Fannie Mae (2023). These trends suggest that household formation and space consumption have become important margins through which rental markets adjust.

Prior research in urban economics and real estate has examined density primarily in geographic terms (e.g., persons-per-square mile) and linked it to wages, rents, and productivity through agglomeration effects Titman et al. (2024); Liu et al. (2018). Our focus differs by emphasizing density at the rental household level, which directly reflects how many people share a given housing unit. We supplement this measure with persons-per-bedroom metrics to capture crowding more precisely and to account for changes in housing composition, including the growth of single-family rentals. Because household-level density can change even when geographic density and occupancy remain stable, it provides a complementary perspective on demand pressures in constrained markets.

Building on standard models of housing demand in which households value space but face budget constraints Muth (1969); Molloy (2022), we test whether excess crowding predicts subsequent rent adjustment. Using panel data for the 100 largest U.S. metropolitan areas, we estimate an error-correction specification in which the next-year's real rent growth responds to excess crowding, short-run changes in household density, and new supply. We find that crowding beyond demographic and structural fundamentals is associated with higher subsequent rent growth, consistent with a stock-flow adjustment process in which prices respond to latent demand pressures over time. Figure 1 illustrates the relationship between excess crowding and future real rent growth.

While the majority of our analysis focuses on excess crowding, we also document empirical relationships between the change in Rental Density Index (RDI) and subsequent rent dynamics. Table 1 shows the near-term effects of densification or expansion, conditioned on starting density. The observed patterns in the level and change of RDI are consistent with the Wheaton-DiPasquale four-quadrant model. In high-density markets, periods of expansion (declines in RDI) reflect the release of pent-up household formation into scarce space, further tightening an already constrained space market

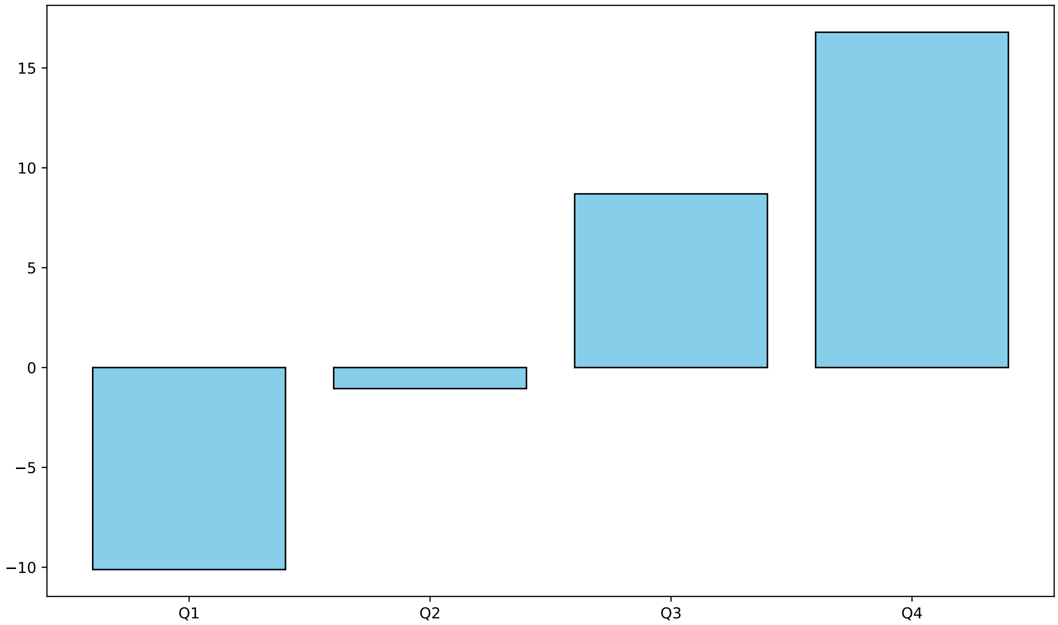

Fig. 1: Difference in means of relative real rent growth: markets with and without Excess Crowding. Excess Crowding is evaluated each year. The rent growth is measured the following year. The data is from the 100 largest Core-Based Statistical Areas (CBSA) for each year between 2005 and 2025.

and leading to strong subsequent rent growth. By contrast, in low-density markets, de-densification occurs in the presence of slack: households expand into underutilized space, increasing effective vacancy and placing downward pressure on rents. Although changes in RDI can arise mechanically through changes in either renter population or renter households, declines in rental unit counts are rare in our sample, occurring in fewer than 50 of 2,000 CBSA-year observations. And we explicitly control for supply changes in our empirical specifications.

Table 1: Rent Outcomes by Density Level and Changes

Initial Rental Density	Density Change Regime	Mean Rent Change (bps)	Observations
High Density	Crowding	23	413
High Density	Expanding	14	587
Low Density	Crowding	-5	537
Low Density	Expanding	-8	463

Notes: Markets are classified by initial rental household density (high vs. low) and by subsequent changes in renter household density (densifying vs. de-densifying). Mean rent variation reflects average next-period real relative rent growth in basis points. The pattern is consistent with a Wheaton-DiPasquale stock-flow adjustment framework: in high-density (tight) markets, de-densification signals the release of pent-up household formation into scarce space and is associated with stronger rent growth, while in low-density (slack) markets, both densification and de-densification are associated with weak or negative rent pressure.

This paper makes three principal contributions. First, we introduce the Rental Density Index (RDI), a household-level measure of rental market tightness constructed from renter population and renter household data. We show how it can be decomposed into: a predictable component driven by demographic structure and unit size; and a residual component capturing excess crowding. Second, we model rental markets as a stock-flow adjustment process and demonstrate that excess crowding functions as an error-correction term: deviations of household density from demographic fundamentals predict subsequent rent growth, conditional on new supply. Finally, we illustrate the practical relevance of this framework through a set of empirical applications that clarify when observed crowding reflects temporary adjustment versus persistent demand pressure, with implications for investors, developers, and policymakers.

The remainder of the paper proceeds as follows. Section 2 reviews related literature on housing demand measurement. Section 3 describes the data and construction of the density measures. Section 4 presents the empirical strategy and main results. Section 5 evaluates forecasting and validation exercises. Section 6 concludes by discussing implications and limitations.

2 Background and Literature Review

A substantial literature in urban and housing economics has sought to measure rental demand using structural modeling, hedonic estimation, and affordability thresholds. Price elasticity estimates, often based on income and housing cost shares, provide insight into how demand responds to price movements under equilibrium assumptions Green et al. (2002); Malpezzi and Green (1996). Hedonic models explain rent levels as a function of unit characteristics and neighborhood amenities, while housing affordability metrics attempt to measure stress via cost-to-income ratios. Though valuable, these approaches often require detailed microdata that are unavailable across time and space or rely on equilibrium assumptions that abstract from short-run adjustment frictions, limiting their usefulness in settings with constrained supply or delayed price response.

A related strand of research emphasizes stock-flow dynamics in housing markets, in which quantities adjust slowly while demand conditions evolve more rapidly Wheaton (1991); Goodman (1992). In this framework, prices need not clear the market instantaneously; instead, adjustment may occur through non-price margins when supply is slow-moving or adjustment costs are high. This perspective is central to understanding rental markets during periods of rapid population growth, migration, or credit-induced shifts in tenure choice, when observed rents may lag underlying demand pressure.

Within applied real estate analysis, the multifamily housing sector has traditionally relied on indicators such as occupancy rates and net absorption to characterize demand. These measures are intuitive and operationally convenient but are tightly linked to the existing housing stock. Occupancy, by definition, cannot exceed 100 percent, and absorption is mechanically bounded by the number of new units delivered Mueller and Laposa (1999); Gabriel and Nothaft (2001). As a result, such metrics are poorly equipped to detect latent demand—periods in which prospective renters are unable to form independent households or must cohabitate due to limited supply or rising costs Sirmans et al. (1991); Pyhrr et al. (1999). This limitation has become more acute in recent years as algorithmic revenue management systems increasingly target stabilized occupancy thresholds, further muting variation and diminishing their informational content Calder-Wang and Kim (2024).

We recognize our supposition may be a gap between academic and emperical veiwpoints. Perhaps large homebuilders rely on other measures of demand, and the limitations of occupancy and absorption do not serve to obfuscate their view of demand. To evaluate this, we interviewed attendees of the 2026 National Multifamily Housing Conference. While these interviews are inherrently anecdotal, they are illustrative. We spoke with Adrian Foley, who is the Chief Executive Officer of Brookfield Residential. This is the homebuilding group within Brookfield, one of the world's largest alternative asset managers, with over USD 1 Trillion under management. Mr. Foley's group focuses on developing single-family homes for two purposes: selling (Build-to-Sell, or BTS) and renting (Build-to-Rent, or BTR). We posed to him the question, "How do you determine the quantity of units to build in a given site or metro?" His response was "It's driven off population growth, jobs growth, and household formation. We overlay that with affordability and you get housing demand by price. It's science-driven but very artsy in execution." We view this response in two ways. First, it suggests that homebuilders do not consider occupancy and absorption to the extent we supposed. However, in their place, they opt for slower-moving more demographic-based metrics. Is it because more housing-based measures do not contain sufficient information, or becasue the other measues – population, jobs, and household growth – contain even more information than occupancy and absorption? A fulsome evaluation of the information contained in those variables is beyond the scope of this investigation. We are affirmed that Mr. Foley did not reference absorption or occupancy explicitly, but instead referenced variables that are unbounded. This does not mean that there is no power in occupancy or absorption, but it does suggest that experts could benefit from a more housing-specific variable that indicates demand without the mechanical constraints of other variables.

Prior research has long recognized that rental market adjustment occurs in part through household formation and crowding behavior. Studies such as Gabriel and Nothaft (2001) and Sirmans et al. (1991) document how constrained rental markets influence living arrangements, vacancy duration, and rent dynamics. However, this literature typically treats household formation outcomes as implicit or secondary margins of adjustment rather than as observable indicators of accumulated market pressure. As a result, changes in household size and crowding are often interpreted as outcomes of tight markets rather than as state variables that summarize the degree of imbalance preceding price adjustment.

Parallel work in urban economics has examined density primarily in geographic terms, such as people or dwellings per square mile, and linked it to productivity, wages, and agglomeration effects Glaeser et al. (2001); Duranton and Puga (2004). More recent studies explore the role of densification in shaping housing affordability and urban form Ahlfeldt and Pietrostefani (2019); Albouy et al. (2015). While informative about land use and spatial efficiency, geographic density does not directly capture household formation or space consumption behavior and may remain stable even as living arrangements adjust within the existing housing stock.

Our work reframes density through a behavioral and dynamic lens by defining it at the rental household level. We introduce the Rental Density Index (RDI), defined as the number of renters per occupied rental unit. Rather than measuring spatial intensity, RDI captures how households adjust their consumption of housing space when demand outpaces supply. In markets where renters prefer more space, increases in RDI reflect household consolidation driven by constrained conditions, while declines indicate de-densification as pressure eases. Crucially, these adjustments can occur even when occupancy remains effectively fixed, allowing RDI to reveal latent demand pressure in fully occupied markets.

We further distinguish between the level of rental crowding and short-run changes in household density. After adjusting RDI for demographic composition and unit size, the residual component—excess crowding—can be interpreted as a slow-moving state variable summarizing accumulated imbalance between population growth and housing supply. By contrast, short-run changes in RDI capture contemporaneous densification that may temporarily absorb pressure without immediately translating into rent growth. This distinction aligns naturally with a stock-flow framework in which supply responds gradually to persistent imbalances rather than to transitory shocks.

RDI's contribution is therefore not simply that it is unbounded, but that it captures a fundamentally different margin of adjustment. Unlike vacancies or absorption, which are constrained by the existing stock, and unlike prices, which may adjust with a lag, household density directly reflects how renters respond to scarcity at prevailing prices. In this sense, RDI provides information about market tightness that is orthogonal to traditional stock-based indicators and complementary to price-based measures.

In sum, our contribution is to reinterpret the familiar concept of density as household density, using a renter-centered indicator that makes household formation and space consumption an explicit part of rental market analysis. By linking excess crowding and short-run densification to subsequent rent growth, we provide a framework that bridges reduced-form forecasting with behavioral interpretation, with practical relevance for investors, planners, and policymakers.

3 Data and Descriptive Statistics

The data used to calculate RDI are from the Public Use Microdata Statistics (PUMS). The PUMS data is an impressive and valuable offering of the U.S. Census in conjunction with the American Community Survey. The data allow for visibility into myriad aspects of a respondent's life (including the number of people in their household and whether they rent or own their home), anonymized for privacy and standardized to create a comprehensive view of the entire country. This data is tabulated by the IPUMS USA Ruggles et al. (2025). The Rental Density Index (RDI), is calculated by summing the number of people living in rental units within a CBSA, and dividing that by the sum of occupied rental units within an CBSA. Excluded from both the numerator and denominator are: group quarters (institutions); vacant units (for sale; for rent; rented or sold but not-yet-occupied); units for seasonal, recreational or other occasional use; and units for migratory workers. We compute its year-over-year difference—ΔRDI—to reflect short-term demand dynamics. This measure avoids the bounded structure of traditional demand indicators like occupancy and absorption, which are upper bounded by 100%.

Also from the Public Use Microdata Statistics are bedrooms-per-household, and age of each person in the household. We calculate bedroom-density by dividing the number of renters in a household by the count of bedrooms in the household. We enrich our CBSA-data with age-range composition figures, by grouping households' residents into bins and converting this to percentage at the CBSA-level.

Our market rent data and inventory growth data is sourced from Costar. This includes nominal effective rent per square foot and inventory units. Our inflation measures come from the U.S. Bureau of Labor Statistics (Series CPIAUCSL). We restrict our analysis to the 100 largest U.S. metropolitan statistical areas (CBSAs) by multifamily inventory as of 2005, ensuring robust sample representation and consistency over time.

To adjust for inflation, we convert nominal rent figures into real terms by subtracting the annual Consumer Price Index change from the annual rent growth change. Although CBSA-level inflation indices excluding rent would be ideal, such data are unavailable at a consistent and granular level. We carefully considered the feasibility of constructing CBSA-level inflation adjustments. We explored both FHFA metropolitan house price indexes and HUD Fair Market Rent series as potential candidates. FHFA indexes measure transaction prices in the owner-occupied housing market and are dominated by single-family sales, making them conceptually mismatched to rental rent growth in multifamily markets. HUD Fair Market Rents, while rental-based, are policy benchmarks designed for voucher administration; they incorporate smoothing, lagged information, and national CPI inputs. They are not intended to measure year-to-year market rent inflation. Because neither source provides a consistent, market-based measure of local rental inflation over our sample period, we instead remove common inflationary effects by using the national CPI.

The Costar data references the same geographic regions as our IPUMS data, as defined by the 2013 Core Based Statistical Areas. Geographic equivalence is an important consideration when joining datasets from different sources, especially when working in a spatio-temporal context like ours. We are assured of equivalency based on the adherence of IPUMS and Costar to the Census-defined TIGERweb Geography files. These files are provided in compliance with the 1998 Section 508 amendment of the Workforce Rehabilitation Act of 1973, § 1194.22 Web-based Intranet and Internet Information and Applications.

To standardize our rent-growth figures, we subtract each year's median rent growth value from each CBSA's value in that year. This gives our rent growth an annual mean near zero. This dampens effects caused by the whipsaw in pricing seen during and after the COVID-19 crisis.

Table 2: Key variables used in the empirical analysis

Variable	Source	Description
`age_under_18_share`	U.S. Census	Share of the CBSA population younger than 18
`age_18_to_24`	U.S. Census	Share of the CBSA population aged 18 to 24
`age_25_to_34`	U.S. Census	Share of the CBSA population aged 25 to 34
`age_35_to_49`	U.S. Census	Share of the CBSA population aged 35 to 49
`age_50_and_over_share`	U.S. Census	Share of the CBSA population aged 50 or older
`bedroom_density_rented`	U.S. Census	Total number of renters in an CBSA divided by the total number of bedrooms in rented households
`rental_households`	U.S. Census	Number of renter-occupied housing units in an CBSA
`rental_population`	U.S. Census	Number of people living in rental dwellings within an CBSA
`CPI`	BLS	Consumer Price Index for All Urban Consumers: All Items, U.S. City Average
`effective_rent_growth_12_m`	CoStar	Effective rent growth in a market over the prior 12 months
`inventory`	CoStar	Total number of rental units in an CBSA
%Δ`CPI`	Calculation	December-to-December percent change in `CPI`
`RDI`	Calculation	`rental_population` divided by `rental_households`
Δ`RDI`	Calculation	Year-over-year change in `RDI`
`real_rent_growth`	Calculation	`effective_rent_growth_12_m` minus %Δ`CPI`
`relative_real_rent_growth`	Calculation	`real_rent_growth` less that year's median `real_rent_growth`
`supply_growth`	Calculation	Year-over-year percentage change in `inventory`

3.1 Rental Density Index Construction

The PUMS data is large, and complex, though very well documented. The output of the raw data requires parsing to aggregate to the CBSA level. This section describes that aggregation process.

3.1.1 Filtering and Labeling

Records are filtered to include only non-vacant units and non-group quarters GQ. This ensures that the analysis captures conventional household formation behavior rather than institutional or non-household living arrangements.

$$\text{VACANCY} = 0 \tag{1}$$
$$\text{GQ} \notin \{3, 4, 5, 6\} \tag{2}$$

Tenure is translated from numeric codes to descriptive labels as follows:

$$\text{OWNERSHP} = \begin{cases} \texttt{"OWNED"} & \text{if code} = 1 \\ \texttt{"RENTED"} & \text{if code} = 2 \\ \texttt{"OTHER"} & \text{otherwise} \end{cases} \tag{3}$$

3.1.2 Grouping Records

Next, the data are aggregated in two stages. At the household level, we sum person weights (PERWT) to obtain household population counts, retain the household weight (HHWT) as the number of households, and compute total bedrooms (BEDROOMS) by weighting unit bedroom counts by household weights.

For each household h, year y, metro area m, and tenure t we compute:

$$\text{POP}_{h,y,m,t} = \sum_{i \in h} \text{PERWT}_i \tag{4}$$
$$\text{HH}_{h,y,m,t} = \text{HHWT}_h \tag{5}$$
$$\text{BEDROOMS}_{h,y,m,t} = \text{BEDROOMS}_h \times \text{HHWT}_h \tag{6}$$

where PERWT_i is the person weight for individual i, and HHWT_h is the household weight taken from the first individual in the group.

These household-level quantities are then aggregated to the metropolitan-year-tenure level, yielding total population, total households, and total bedrooms by ownership status within each metropolitan area and year.

$$\text{POP}_{y,m,t} = \sum_{h \in (y,m,t)} \text{POP}_{h,y,m,t} \tag{7}$$
$$\text{HH}_{y,m,t} = \sum_{h \in (y,m,t)} \text{HH}_{h,y,m,t} \tag{8}$$
$$\text{BEDROOMS}_{y,m,t} = \sum_{h \in (y,m,t)} \text{BEDROOMS}_{h,y,m,t} \tag{9}$$

3.1.3 Feature Construction

Densities by year, metro, and tenure are calculated at the household and bedroom level.

$$\text{DENSITY}_{y,m,t} = \frac{\text{POP}_{y,m,t}}{\text{HH}_{y,m,t}} \tag{10}$$

$$\text{BEDROOM\_DENSITY}_{y,m,\text{rent}} = \frac{\text{POP}_{y,m,\text{rent}}}{\text{BEDROOMS}_{y,m,\text{rent}}} \tag{11}$$

$$\text{RDI}_{y,m} \equiv \text{DENSITY}_{y,m,t=2} \tag{12}$$

$$\Delta\text{RDI}_{y,m} = \text{RDI}_{y,m} - \text{RDI}_{y-1,m} \tag{13}$$

Age cohorts are calculated at the metro-year level, for the renter tenure, according to the following brackets:

$$\begin{aligned} g_{<18} &: \ 0 \leq \text{AGE} \leq 17 \\ g_{\text{18-24}} &: \ 18 \leq \text{AGE} \leq 24 \\ g_{\text{25-34}} &: \ 25 \leq \text{AGE} \leq 34 \\ g_{\text{35-49}} &: \ 35 \leq \text{AGE} \leq 49 \\ g_{50+} &: \ \text{AGE} \geq 50 \end{aligned}$$

using the following method:

$$\text{AGE\_SHARE}_{y,m,g} = \frac{\sum_{i \in (y,m,r)} \mathbb{1}\{\text{AGE}_i \in g\}\, \text{PERWT}_i}{\text{TOTAL\_RENTERS}_{y,m}} \tag{14}$$

We connect the features above to the Costar records of inventory and rent growth by joining on IPUMS' MET2013 to Costar's CBSA_Code:

3.2 Summary Statistics

Table 3 provides descriptive statistics for the key variables across the full panel. On average, the `density_rented` (RDI) across all CBSAs and years is 2.4 persons per rental unit, with a standard deviation of 0.3. The average annual growth in rental supply is 2.0%, while the average `rent_growth` is 2.5%, when we remove the effects of inflation and normalize the value to the median rent growth of the year, the `real_relative_rent_growth_next_year` drops to a mean of 0.10%. The `bedroom_density` (people-per-bedroom) in rental households is 0.12 higher than in the owned-home counterpart. The missing records exist only in the beginning or ending years as the prior period and next-period values do not exist. The latest date for which Census data exists is 2024, while the latest full-year of rent data is from 2025.

3.3 Coverage and Representativeness

The sample includes 2,000 CBSA-year observations, covering a mix of large coastal cities, fast-growing Sun Belt metros, and slower-growing Midwestern regions. Together, these CBSAs account for over 32 million rental dwellings, housing over 73 million residents, providing a true representative snapshot of national rental dynamics. In

Table 3: Summary statistics

	N	mean	std	min	Median	max
year	2,000	2,014	5.77	2,005	2,014	2,024
inventory_units	2,000	127,489	189,877	14,696	54,792	1,457,169
supply_growth	1,900	0.02	0.02	-0.02	0.01	0.16
rent_growth	2,000	0.02	0.03	-0.16	0.02	0.32
real_relative_rent_growth_next_year	2,000	0.00	0.02	-0.12	0.00	0.28
population_owned	2,000	1,311,841	1,592,831	115,542	729,340	11,420,211
population_rented	2,000	704,440	1,127,270	76,565	323,210	9,041,819
households_owned	2,000	473,933	549,249	47,432	273,765	4,063,756
households_rented	2,000	286,193	438,032	38,095	136,014	3,744,508
density_owned	2,000	2.70	0.20	2.18	2.66	3.38
density_rented	2,000	2.40	0.28	1.77	2.35	3.40
density_rented_change	1,900	-0.01	0.09	-0.40	-0.00	0.41
bedroom_density_rented	2,000	0.78	0.08	0.60	0.77	1.09
bedroom_density_owned	2,000	0.64	0.04	0.54	0.63	0.82
age_under_18_share	2,000	0.26	0.04	0.12	0.26	0.40
age_18_to_24_share	2,000	0.14	0.04	0.07	0.13	0.43
age_25_to_34_share	2,000	0.21	0.02	0.14	0.20	0.31
age_34_to_49_share	2,000	0.21	0.02	0.10	0.21	0.27
age_50_and_over_share	2,000	0.20	0.04	0.10	0.20	0.34

our Emperical Strategy and Results, we evaluate the degrees to which our model specification varies when applied to sparesly populated areas, outside of our coverage.

3.4 Preliminary Observations

Across the 100 CBSAs in the study, the RDI ranges from 1.77 (2020: Albany-Schenectady-Troy, NY) to 3.70 (2012: Riverside-San Bernardino-Ontario, CA). This wide range reflects both housing market constraints and regional variation in family formation and living preferences. The RDI level alone cannot distinguish whether units are full by necessity (housing shortage) or by choice, so we complement it with additional measures: renter age distributions to account for lifecycle preferences; and bedroom-density metrics for both rented and owned households to measure actual crowding net of compositional shifts. In the next section we investigate how these measures interact. Recognizing younger populations' higher density tolerance and the growth in single-family rentals, we expect these factors to shape observed crowding patterns, and we aim to identify excess demand beyond what demographic composition alone would predict.

4 Empirical Strategy and Results

4.1 Measuring Rental Crowding and Demographic Fundamentals

We begin by modeling rental household density as a function of demographic composition and unit characteristics in order to isolate a measure of crowding that is not mechanically driven by observable fundamentals. Specifically, we estimate the following panel regression for metropolitan area m in year t:

$$\begin{aligned}\text{Density}_{mt} = \alpha_m + \gamma_t + \beta_1\,\text{BedroomDensity}_{mt} + \beta_2\,\text{AgeShare}_{<18,mt} \\ + \beta_3\,\text{AgeShare}_{25\text{–}34,mt} + \beta_4\,\text{AgeShare}_{35\text{–}49,mt} + \varepsilon_{mt}.\end{aligned} \tag{15}$$

where α_m denotes metropolitan fixed effects, and γ_t denotes year fixed effects.

Density_{mt} is defined as renters per rental household. The specification includes metropolitan and year fixed effects to absorb time-invariant spatial heterogeneity and common macroeconomic shocks. Standard errors are clustered at the metropolitan level. Table 4 reports the results.

Table 4: Determinants of Rental Household Density

	(1) Rental household density
Bedroom density (renters per bedroom)	2.4471***
	(0.0753)
Age share under 18	1.0310***
	(0.1237)
Age share 25–34	-0.2578**
	(0.0917)
Age share 35–49	-0.0064
	(0.1026)
Constant	0.2631***
	(0.0496)
Metropolitan fixed effects	Yes
Year fixed effects	Yes
Observations	2,000
R^2_{within}	0.854

Notes: The dependent variable is rental household density, defined as renters per occupied rental unit. All specifications include metropolitan and year fixed effects. Standard errors clustered at the metropolitan level are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Importantly, our objective is not to identify a causal effect of crowding on rents. Rather, we evaluate whether renter household density functions as a measurable state variable that summarizes latent market pressure arising from slow supply adjustment and household formation responses. Throughout the paper, we therefore emphasize predictive content and market interpretation rather than causal attribution.

The model explains a substantial share of within- and between-metro variation in rental density, with an overall R^2 of 0.86 and a withinR^2 of 0.85. Bedroom density is the dominant predictor of rental crowding: a one-unit increase in renters per bedroom is associated with a 2.45 increase in renters per rental household, reflecting the mechanical relationship between unit size and observed density.

Demographic composition also plays an economically meaningful role. Metropolitan areas with a higher share of children exhibit significantly greater rental density, consistent with larger household sizes among renter families. In contrast, a higher share of residents aged 25-34 is associated with lower rental density, consistent with a greater prevalence of single-person or roommate households among young adults. The coefficient on the 35-49 age group is small and statistically insignificant once fixed effects are included.

We define excess crowding as the residual from this regression. It is the component of rental density not explained by demographics, unit size, or persistent metropolitan characteristics. This residual is interpreted as a state variable capturing latent demand pressure in the rental market. We parameterize a regression using excess crowding and observed supply growth to explain next-period real relative rent growth. Results are reported in Table 5.

Because our framework emphasizes rent adjustment in response to imbalances between latent demand pressure and supply response, we also consider an alternative formulation that treats both sides of the market symmetrically. In addition to *excess* crowding, we construct a measure of *excess* supply growth, defined as metropolitan-level rental inventory growth net of the national mean in a given year. This transformation isolates cross-metropolitan variation in supply response, abstracting from common cyclical conditions and aggregate construction booms or slowdowns absorbed by year fixed effects.

We then estimate an alternative rent-adjustment specification in which next-year relative real rent growth responds jointly to excess crowding and excess supply growth. The results are reported in Table 6.

The results of both specifications are consistent with the stock-flow adjustment interpretation. Excess crowding remains a positive and statistically significant predictor of subsequent rent growth, while excess supply growth enters with a negative and economically meaningful coefficient. Markets that experience supply growth above the national average exhibit slower relative rent adjustment, conditional on latent demand pressure.

This second specification does not aim to identify causal supply elasticities. Rather, it sharpens the interpretation of the rent-adjustment equation as reflecting relative imbalance: prices adjust most strongly in markets where latent demand pressure is high and local supply response is weak, both measured relative to contemporaneous national conditions. The stability of the excess crowding coefficient in this formulation reinforces its role as a summary state variable capturing accumulated market pressure rather than transitory fluctuations or compositional shifts.

The overall explanatory power of both specifications is modest, which is typical for one-period-ahead rent growth regressions with metropolitan and year fixed effects. Importantly, our objective is not to explain the level of rent growth, but to assess whether excess crowding contains incremental predictive information about cross-metropolitan rent adjustment. When omitting the pandemic year (2020), the R^2 increases to 0.0591; additionally omitting the peak year of the Global Financial Crisis (2008) renders an R^2 of 0.0727. This is consistent with the view that excess crowding summarizes medium-run imbalances whose predictive content is obscured during periods of extreme disruption.

Table 5: Excess Crowding and Next-Year Relative Real Rent Growth

	(1) Relative real rent growth, t+1
Excess crowding	0.0367**
	(0.0144)
Supply growth	-0.2135***
	(0.0449)
Constant	0.0046***
	(0.0009)
Metropolitan FE	Yes
Year FE	Yes
Observations	1,900
R^2_{within}	0.028

Notes: The dependent variable is real, relative rent growth in year t+1. All specifications include metropolitan and year fixed effects. Standard errors clustered at the metropolitan level are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 6: Excess Crowding, Excess Supply Growth, and Next-Year Relative Real Rent Growth

	(1) Relative real rent growth, $t + 1$
Excess crowding	0.0300**
	(0.0149)
Excess supply growth	-0.2143***
	(0.0449)
Constant	0.0004***
	(0.0000)
Metropolitan fixed effects	Yes
Year fixed effects	Yes
Observations	1,900
R^2 (within)	0.031

Notes: The dependent variable is real, relative rent growth in year $t + 1$. Excess supply growth is defined as metropolitan rental inventory growth net of the national mean in a given year. All specifications include metropolitan and year fixed effects. Standard errors clustered at the metropolitan level are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

4.2 A Stock-Flow Interpretation of Excess Crowding

This section provides a simple stock-flow motivation for the empirical specification used in the paper. The objective is not to develop a fully structural model of rental markets, but rather to clarify why deviations in renter household density from fundamentals—excess crowding—naturally enter a rent-adjustment equation in a setting with slow-moving supply.

Let $D_{m,t}$ denote renter household density in metropolitan area m at time t, defined as renters per occupied rental unit. Households have a preferred or long-run density level $D^*_{m,t}$, which depends on demographic composition and housing characteristics such as unit size:

$$D^*_{m,t} = f(X_{m,t}),$$

where $X_{m,t}$ includes age composition and bedroom density. When housing supply adjusts slowly, actual density may deviate from this benchmark.

Define excess crowding as the deviation of observed density from its long-run fundamental level:

$$EC_{m,t} = D_{m,t} - D^*_{m,t}.$$

Positive values of $EC_{m,t}$ reflect household consolidation arising when demand pressure is absorbed through space consumption rather than immediate price adjustment.

In a stock-flow framework, rents respond gradually to accumulated imbalance rather than instantaneously clearing the market. A simple partial-adjustment relationship for real rent growth can be written as:

$$\Delta R_{m,t+1} = \lambda\, EC_{m,t} - \gamma\, \Delta S_{m,t} + \varepsilon_{m,t+1},$$

where $\Delta R_{m,t+1}$ denotes next-period real rent growth, $\Delta S_{m,t}$ is new supply growth, and $\lambda > 0$ captures the speed at which excess crowding is translated into rent adjustment.

This formulation highlights two features of excess crowding. First, it is a state variable: it summarizes the cumulative effect of past demand pressure that has not yet been fully reflected in prices. Second, its predictive content is inherently forward-looking: excess crowding affects future rent growth because it reflects imbalances that are resolved only gradually as supply and prices adjust.

The empirical specification in Section 4.1 corresponds directly to this reduced-form adjustment equation. Excess crowding is constructed as the residual from a regression of renter household density on demographic and structural fundamentals, while supply growth is included to capture contemporaneous quantity adjustment. The presence of metropolitan and year fixed effects absorbs time-invariant heterogeneity and common macroeconomic shocks, leaving identification to rely on cross-metropolitan variation in accumulated imbalance.

Importantly, this framework does not imply that excess crowding causes rent growth in a structural sense. Rather, it formalizes why excess crowding contains information about future rent adjustment in markets where supply responds slowly and households adjust along non-price margins.

4.3 Dynamic Ordering Between Excess Crowding and Rent Growth

To further assess the interpretation of excess crowding as a state variable in a stock-flow adjustment process, we examine the dynamic ordering between renter household density and rent growth. To do this we evaluate Granger Causality. We emphasize that this analysis does not establish structural causality. Instead, it evaluates whether excess crowding contains predictive information for future rent adjustment beyond that already embedded in past rents, and whether rent growth predicts subsequent changes in crowding.

Specifically, we estimate two panel regressions with metropolitan and year fixed effects. The first relates next-year real relative rent growth to contemporaneous excess crowding and lagged rent growth. The second examines whether rent growth predicts subsequent excess crowding after controlling for the persistence of crowding itself. Standard errors are clustered at the metropolitan level.

The results are reported in Tables B6 and B7. Consistent with a stock-flow adjustment framework, excess crowding predicts subsequent rent growth even after conditioning on lagged rents. By contrast, lagged rent growth has no predictive power for future excess crowding, while excess crowding itself is highly persistent. This asymmetric dynamic relationship supports the interpretation of excess crowding as a slow-moving state variable that precedes price adjustment, rather than as an outcome driven mechanically by recent rent changes.

4.4 Robustness Checks

We assess robustness along four dimensions: outcome definition, assignment of excess crowding, sensitivity to slow-moving metropolitan trends, and heterogeneity in the rent-adjustment response across markets with differing baseline density.

4.4.1 Placebo tests.

To evaluate whether the baseline fixed-effects results reflect mechanical correlations rather than economically meaningful relationships, we implement two placebo tests. First, we replace the primary outcome with nominal rent growth. Second, we randomly reassign excess crowding across metropolitan areas within each year. In both cases, the estimated effects collapse in magnitude or center tightly at zero, indicating that the baseline relationship is not driven by general inflation or spurious correlation. Results are reported in Appendix B.

When excess crowding is randomly reassigned across metropolitan areas within year, the resulting coefficient distribution is tightly centered at zero (mean -0.00006; 95% empirical interval $[-0.028, 0.026]$) and does not approach the magnitude observed in the baseline specification. Details are reported in Table B3.

4.4.2 Nominal rent growth.

Excess crowding exhibits limited predictive power for nominal rent growth, with an effect size substantially smaller than that observed for real relative rent growth. This finding indicates that the main results are not driven by common inflationary forces or mechanical price momentum. Results are reported in Table B2.

4.4.3 Sensitivity to slow-moving metropolitan trends.

Because renter household density is a slow-moving variable, excess crowding may partially reflect medium-run metropolitan adjustment paths rather than purely transitory deviations. To assess sensitivity to this concern, we estimate specifications that allow for metropolitan-specific linear time trends using the Frisch-Waugh-Lovell theorem. Specifically, for each metropolitan area we regress both the outcome and excess crowding on a constant and time, retain the residuals, and then estimate the standard two-way fixed-effects regression on those residuals. This procedure is algebraically equivalent to including metropolitan-by-time trends.

When estimated on the full sample, including the pandemic year (2020), allowing for metropolitan-specific trends attenuates the estimated short-run relationship between excess crowding and subsequent rent growth (Table B4). However, excluding 2020 restores statistical significance (Table B5), suggesting that the attenuation in the full sample is driven by the exceptional disruptions of the pandemic rather than by long-run growth trajectories. Because excess crowding is intended to summarize accumulated imbalance over time, specifications with metropolitan-specific trends are intentionally conservative.

4.4.4 Heterogeneity by baseline rental density.

Finally, we examine whether the rent-adjustment response to excess crowding varies across markets with differing baseline utilization of housing space. The interpretation of renter household density as a measure of latent demand pressure relies on space consumption being a binding margin of adjustment, which need not hold uniformly across markets. We therefore split the sample based on baseline rental bedroom density and re-estimate the rent-adjustment equation separately for high- and low-density metropolitan areas, holding the construction of excess crowding fixed across samples.

Results are reported in Table 7. In metropolitan areas with above-median bedroom density, excess crowding is a positive and statistically significant predictor of subsequent rent growth, while the effect of supply growth is weaker. In contrast, in below-median density markets excess crowding has no predictive power, and rent adjustment is driven primarily by variation in supply growth. These patterns are consistent with the stock-flow adjustment framework: in dense, space-constrained markets, latent demand pressure is absorbed through household consolidation before prices adjust, whereas in lower-density markets adjustment occurs more directly through physical supply expansion or vacancy changes.

4.5 Discussion of Results: Excess Crowding as a Stock-Flow State Variable

The empirical results above motivate a careful interpretation of excess crowding and its role in rental market adjustment. Excess crowding is not intended to measure unmet housing demand in a welfare or Marshallian sense, nor does it represent a structural demand shifter. Rather, it is a reduced-form state variable that summarizes accumulated imbalance between household formation and housing supply in a market where physical adjustment occurs slowly.

Within a stock-flow framework of housing markets, prices do not adjust instantaneously to demand shocks when supply is constrained or adjustment costs are high. Instead, part of the adjustment occurs through non-price margins. For rental housing,

Table 7: Heterogeneity in Rent Adjustment by Baseline Rental Bedroom Density

	(1) High bedroom density	(2) Low bedroom density
Excess crowding	0.0527**	-0.0028
	(0.0261)	(0.0179)
Supply growth	-0.1296*	-0.1942***
	(0.0739)	(0.0684)
Metropolitan fixed effects	Yes	Yes
Year fixed effects	Yes	Yes
Observations	965	935
R^2 (within)	0.019	0.010

Notes: The dependent variable is real, relative rent growth in year $t+1$. Excess crowding is defined as the residual from the baseline rental density regression estimated on the full sample. Bedroom-density groups are defined using the median rental bedroom density measured at baseline. All specifications include metropolitan and year fixed effects. Standard errors clustered at the metropolitan level are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

one such margin is household space consumption. When population growth, migration, or delayed homeownership increases pressure on the rental sector faster than new units can be delivered, households respond by consolidating living arrangements: they may add roommates, delaying household formation, or remain in higher-density units longer than they otherwise would. These responses increase renter household density without immediately clearing the market through prices.

Excess crowding captures this accumulation of pressure. Because it is constructed as the component of renter household density unexplained by demographics, unit size, and persistent metropolitan characteristics, it reflects departures from a market's long-run density baseline rather than slow-moving structural differences across cities. As such, excess crowding functions analogously to an error-correction term in a stock-flow adjustment model: it summarizes past imbalance and predicts subsequent price adjustment as supply and prices gradually respond.

This interpretation clarifies several features of the empirical results. First, the modest explanatory power of one-period-ahead rent growth regressions is expected. Short-run rent changes are influenced by idiosyncratic shocks, measurement noise, and institutional factors such as lease timing. Excess crowding is not intended to explain contemporaneous rent movements, but rather to indicate whether underlying pressure exists that has yet to be fully reflected in prices. Second, the attenuation of the excess crowding coefficient in specifications that include metropolitan-specific linear trends reflects the fact that excess crowding is a slow-moving stock. Removing medium-run adjustment paths necessarily removes part of the signal that excess crowding is designed to capture.

The heterogeneity results further support this interpretation. Excess crowding predicts subsequent rent growth in markets with high baseline bedroom density—where space consumption is a binding margin of adjustment—but not in lower-density markets, where households can adjust more readily through vacancy or physical expansion.

This pattern is consistent with a stock-flow view in which household consolidation absorbs demand pressure only when spatial slack is limited.

Importantly, excess crowding is distinct from commonly used demand indicators such as occupancy and absorption. Occupancy rates are mechanically capped and increasingly stabilized by revenue management practices, while absorption is bounded by the pace of new supply delivery. Both measures are therefore limited in their ability to detect latent pressure in fully occupied or supply-constrained markets. Excess crowding instead reveals whether adjustment is occurring through household behavior rather than prices, providing information about market tightness that prices and stock-based metrics alone cannot convey.

In this sense, excess crowding should be understood as a diagnostic indicator of market imbalance rather than a causal determinant of rents or a measure of housing welfare. Its value lies in identifying when and where rental markets are absorbing demand pressure through consolidation rather than immediate price adjustment, and in forecasting the subsequent release of that pressure into rents as adjustment continues.

5 Value Added by the Rental Density Index in Forecasting Rent Growth

This section evaluates whether the Rental Density Index (RDI) provides incremental predictive information about future rent growth beyond commonly used indicators of rental market conditions, including vacancies, absorption, supply growth, and lagged rent growth. In contrast to the regression results presented earlier, which establish in-sample relationships consistent with a stock–flow adjustment framework, the analysis here focuses explicitly on forecast performance. All exercises are designed to be information-set consistent and to mirror how market participants would form expectations in real time.

We present three complementary exhibits, all out-of-sample. The first evaluates short-term, one-period-ahead predictive power in a direct horse race against standard indicators. The second examines whether persistent changes in RDI contain information about medium-term rent performance. The third provides an internal consistency check, relating the sign and magnitude of RDI changes to subsequent rent outcomes using only variation within the RDI measure itself.

5.1 Short-Term Horse Race: One-Period-Ahead Forecasts

We begin by assessing the short-term predictive power of RDI relative to alternative demand indicators. Using rolling OLS regressions, we generate one-period-ahead forecasts of real relative rent growth for each metropolitan area. In each rolling window, the forecasting model is estimated using only data available up to that point in time. We then evaluate each model's ability to sort markets into future outperformers and underperformers. In this section we refer to "outperform/er/ance" as having excess real relative rent growth. Conversely "underperform/er/ance" refers to a market or markets with lower-than-average real rent growth.

Rather than emphasizing coefficient magnitudes, we focus on ranking performance, which is both robust to scale differences across predictors and directly relevant for investors and policymakers. Specifically, for each forecast year we rank metropolitan areas by predicted rent growth and compare realized outcomes between the top and bottom quintiles, as well as between the top and bottom deciles.

The results are reported in Figure 2. Forecasts based on RDI combined with lagged rent growth consistently outperform those based on any single alternative indicator, including vacancies, absorption, supply growth, or lagged rents alone. Moreover, the RDI-based forecast also outperforms a specification that includes all four alternative indicators jointly. This pattern holds across both quintile and decile comparisons, indicating that the result is not driven by extreme observations.

These findings suggest that RDI captures a dimension of latent demand pressure that is not subsumed by traditional stock-based or flow-based measures, even when those measures are used together. In the short run, household crowding appears to summarize information about market tightness that is otherwise only weakly reflected in vacancies, absorption, or recent price movements.

5.2 Medium-Term Predictive Power and Persistence

While the previous exercise focuses on short-run rent adjustment, RDI is conceptually a slow-moving state variable reflecting accumulated imbalances between household formation and housing supply. We therefore examine whether persistent changes in RDI contain information about medium-term rent performance.

To do so, we compute, for each metropolitan area, the trailing five-year average of the sign of annual RDI changes. For example, if Los Angeles experienced four years of negative RDI changes and one year of positive RDI change, then its value would be 0.6 (the average of -1,-1,-1,-1,1). This measure captures whether a market has been experiencing sustained densification or de-densification, abstracting from short-term volatility. Metropolitan areas are then sorted into two groups: those with above-median and below-median values. We examine those groups mean real relative rent growth over the next five years.

Figure 4 reports the results of this exercise and compares the performance of the RDI-based grouping to analogous groupings constructed using the same strategy applied to vacancies, absorption, supply growth, and lagged rent growth. Using the identical grouping rule across variables, the RDI-based measure produces a clearer and more monotonic separation between future outperformers and underperformers than any of the alternatives.

This evidence indicates that sustained changes in renter household density contain information about medium-term rent dynamics that is not captured by more transitory indicators. Persistent crowding appears to reflect cumulative demand pressure that is released only gradually into prices, consistent with a stock-flow adjustment process in which supply responds slowly to demand shocks.

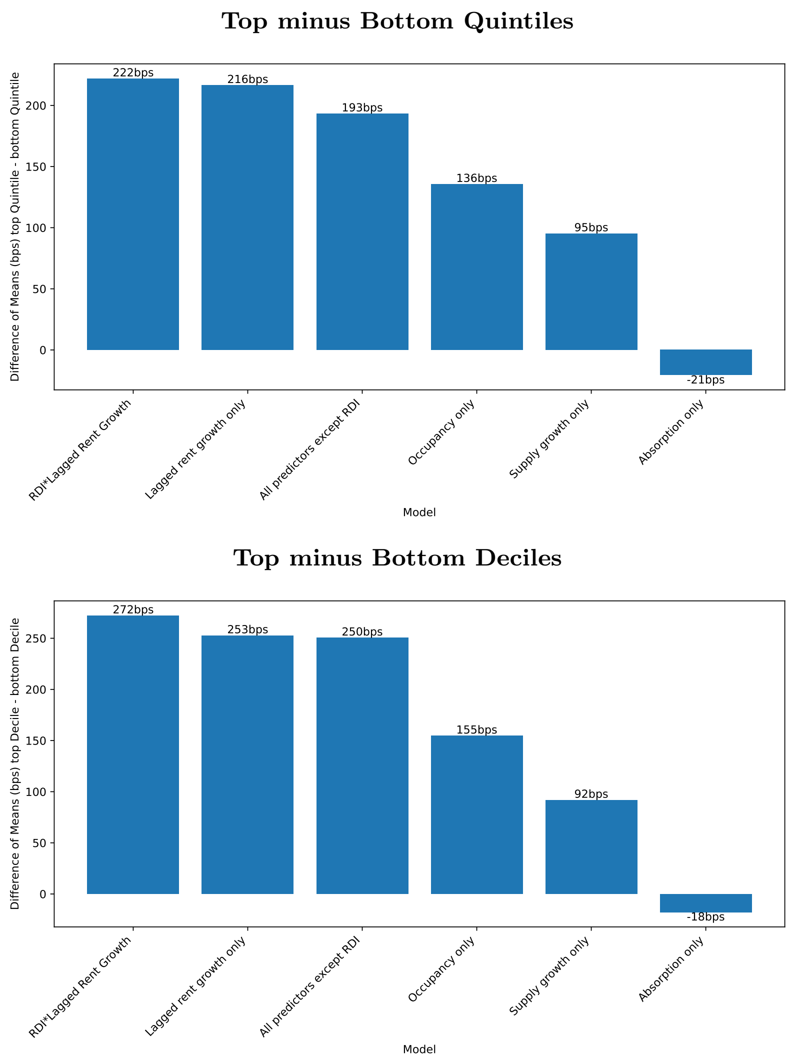

Fig. 2: Short-Term Forecast Horse Race. Difference of means: next-period real relative rent growth in forecasted high-growth and low-growth markets. Markets are ranked each period using rolling OLS forecasts based on individual predictors and selected combinations. Bars report average differences between the top and bottom quintiles (top panel) and top and bottom deciles (bottom panel). The RDI-based forecast (RDI plus lagged rent growth) outperforms all single-variable forecasts and a specification including vacancies, absorption, supply growth, and lagged rent growth jointly.

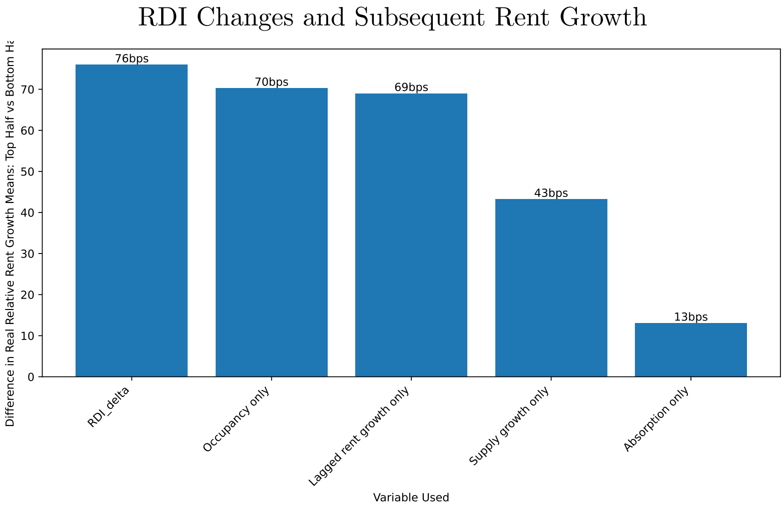

Fig. 3: Average real relative rent growth over the subsequent five years for metropolitan areas grouped by their trailing five-year value of various metrics. We split markets into two groups based on their past RDI, average occupancy, average rent growth, etc. We then take the average real relative rent growth of those groups and plot the differences. The study is performed out-of-sampple, with readings taken in 2005-2009 (2010-2014, 2015-2020) and rent reported for 2010-2014 (2015-2019, 2020-2024)

5.3 Internal Consistency: RDI Changes and Future Rent Growth

Finally, we examine the internal consistency of the RDI signal by relating changes in RDI to subsequent rent outcomes using only variation within the RDI measure itself. This exercise is not intended as a horse race against other predictors, but rather as a credibility check on whether RDI behaves in a manner consistent with its interpretation as a measure of latent demand pressure.

Specifically, we present a histogram where bins are determined based on the sum of the sign of RDI changes over a trailing five-year window. We then examine average rent growth over the subsequent five years. The results, shown in Figure 3, reveal a clear monotonic relationship: markets with more persistently positive RDI changes subsequently experience higher average rent growth, while markets with sustained declines in RDI experience weaker rent performance.

Importantly, this relationship is evaluated out of sample, with grouping based solely on past RDI changes and outcomes measured in future periods. The monotonic pattern reinforces the interpretation of RDI as a meaningful summary of underlying demand pressure rather than a noisy or purely contemporaneous statistic.

Medium-Term Rent Performance by Persistent RDI Changes

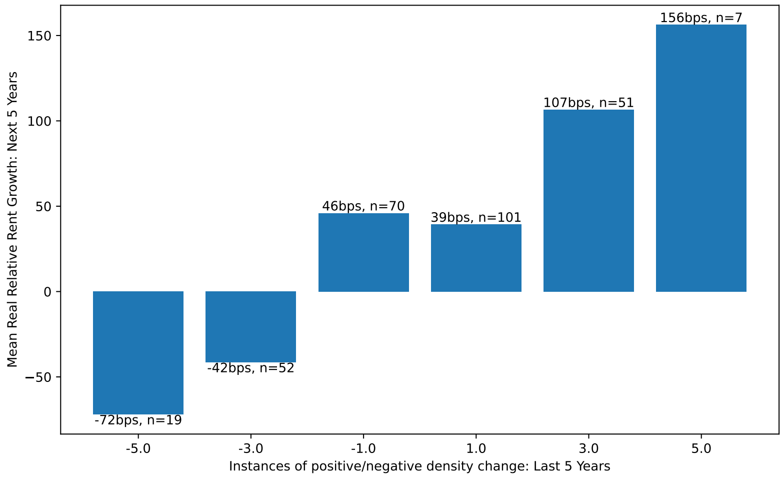

Fig. 4: Average real relative rent growth over the subsequent five years for metropolitan areas grouped by their trailing five-year sum of RDI-change sign. For example, a market that experienced five years of negative RDI growth would be represented in the leftmost bin. That market's next five years of real relative rent growth is summed then averaged with its cohort (other markets who saw five consecutive periods of negative RDI growth). The study is performed out-of-sampple, with RDI readings taken in 2005-2009 (2010-2014, 2015-2020) and rent reported for 2010-2014 (2015-2019, 2020-2024)

5.4 Summary

Taken together, these three exercises demonstrate that RDI provides incremental predictive content beyond standard indicators of rental market conditions. RDI improves short-term forecast performance in direct horse races, captures medium-term dynamics associated with persistent market imbalance, and exhibits internally consistent relationships with future rent outcomes. These findings support the interpretation of renter household density as a behavioral margin through which latent demand pressure accumulates and is ultimately reflected in rental prices. While excess crowding is informative for forecasting and market characterization, it should not be interpreted as a structural measure of demand or welfare without additional identifying assumptions.

6 Conclusion

This paper introduces the Rental Density Index (RDI), a household-level measure of rental market tightness that captures how renters adjust their consumption of housing space when supply responds slowly to changes in demand. By reframing density as the number of occupants per renter household rather than as a geographic concept, we highlight household consolidation as a key non-price margin through which rental markets absorb pressure prior to price adjustment.

Using panel data for the 100 largest U.S. metropolitan areas, we show that deviations of renter household density from demographic and unit-size fundamentals—excess crowding—contain predictive information about subsequent real rent growth, even after conditioning on new supply and standard demand indicators. Interpreted within a stock-flow adjustment framework, excess crowding functions as a slow-moving state variable summarizing accumulated imbalance between household formation and housing supply. Markets with greater excess crowding experience stronger subsequent rent adjustment as prices and quantities gradually respond.

Several limitations of this approach warrant explicit discussion. First, RDI is most informative in markets where space consumption is a binding margin of adjustment. Our heterogeneity results show that excess crowding predicts rent growth primarily in dense, space-constrained metropolitan areas. In lower-density markets, where households can adjust more readily through vacancies, migration, or physical expansion, household density contains little incremental information beyond supply growth. Second, RDI is a medium-run indicator by construction. Its predictive content weakens during periods of extreme disruption—such as the Global Financial Crisis or the COVID-19 pandemic— when institutional shocks, policy interventions, and abrupt behavioral changes dominate normal adjustment dynamics. Finally, RDI abstracts from within-market heterogeneity: it summarizes average household density and does not capture distributional differences across income groups, neighborhoods, or housing types.

The results also clarify the scope of causal interpretation. Excess crowding should not be interpreted as a causal determinant of rent growth, nor as a structural measure of housing demand or welfare. Rather, it is a reduced-form diagnostic that summarizes how markets absorb demand pressure when supply adjusts slowly. Moving from prediction to causal inference would require exogenous variation that shifts household formation or crowding independently of expected rent dynamics—for example, plausibly exogenous migration shocks, policy-induced changes in household eligibility or living arrangements, or instruments that affect household consolidation without directly affecting rental supply or pricing. Such identification strategies lie beyond the scope of this paper but represent a promising direction for future research.

Despite these limitations, the findings have substantive implications for policy and market analysis. For policymakers, rising renter household density may signal accumulating pressure even in markets where rents appear temporarily stable or vacancies remain low. Monitoring household density alongside prices and vacancies can help distinguish whether affordability pressures are being absorbed through consolidation rather than resolved through new supply. This distinction is particularly relevant for evaluating the short-run effects of zoning reforms, construction pipelines, or demand-side subsidies, which may influence household formation behavior before they affect observed rents.

For investors and market participants, measures of renter household density provide a complementary indicator of market tightness that can improve assessment of future rent risk, particularly in supply-constrained and stabilized markets where traditional demand metrics are muted. More broadly, the results underscore that how households share space matters for understanding rental market dynamics.

Overall, this paper shows that household density is not merely a descriptive statistic but a meaningful component of housing market adjustment. By incorporating renter household density into a stock-flow view of rental markets, we provide a new lens for measuring latent demand pressure and anticipating rent movements in constrained environments. Future work that combines household-level data with exogenous shocks or policy variation may further clarify the causal mechanisms underlying these adjustment processes and extend the framework to additional settings.

Acknowledgments

Declarations

Data and Materials Availability Data that comes from public sources is available upon request. Code is also available on request. Rent-growth figures come from Costar, and thus require their permission for redistribution.
Authors' Contributions All authors contributed equally
Funding This paper has not been funded by any organizations
Ethical Approval No humans or animals are involved
Consent to Participate No humans or animals are involved
Consent to Publish No humans or animals are involved
Competing Interests There are no competing interest

Appendix A Shortage Estimates

Table A1: Selected Estimates of the U.S. Housing Unit Shortage

Year	Source	Estimated Housing Shortage (Millions of Units)
2022	Corinth and Dante (2022)	20.1
2024	National Low Income Housing Coalition (2024)	7.1
2020	National Association of Realtors (2020)	5.5
2024	Brookings Institution (2024)	4.9
2025	Zillow (2025)	4.7
2022	Zillow (2022)	4.5
2021	Up for Growth (2021)	3.9
2025	Goldman Sachs (2025), price-to-income approach	3.9
2020	Freddie Mac (2020)	3.8
2024	Freddie Mac (2024)	3.7
2025	Goldman Sachs (2025), vacancy-based approach	3.1
2023	Jones (2023)	2.5
2025	Zandi et al. (2025)	2.0
2022	National Association of Home Builders (2022)	1.5

Notes: Estimates are drawn from a range of academic, industry, and policy sources and reflect differing methodologies, definitions of equilibrium, and time horizons. Across the studies listed, the mean estimated housing shortfall is approximately 5.1 million units and the median estimate is approximately 3.9 million units.

Appendix B Robustness Checks

Table B2: Placebo Test: Excess Crowding and Nominal Rent Growth

	(1) Nominal rent growth, $t+1$
Excess crowding	0.0106
	(0.0400)
Metropolitan fixed effects	Yes
Year fixed effects	Yes
Observations	213
R^2 (within)	0.000

Notes: The dependent variable is nominal rent growth in year $t+1$. Excess crowding is defined as the residual from the baseline rental density regression. All specifications include metropolitan and year fixed effects. Standard errors clustered at the metropolitan level are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table B3: Placebo Test: Randomized Excess Crowding

	(1) Relative real rent growth, t+1
Randomized excess crowding	-0.0112
	(0.0140)
Metropolitan fixed effects	Yes
Year fixed effects	Yes
Observations	2,000
R^2_{within}	0.000

Notes: Excess crowding is randomly reassigned across metropolitan areas within each year. The dependent variable is real, relative rent growth in year t+1. All specifications include metropolitan and year fixed effects. Standard errors clustered at the metropolitan level are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table B4: Placebo Test with Metropolitan-Specific Linear Trends

	(1) Real relative rent growth, $t+1$
Excess crowding	0.0185
	(0.0170)
Metropolitan fixed effects	Yes
Year fixed effects	Yes
Metropolitan-specific linear trends	Yes
Observations	1,000
R^2 (within)	0.002

Notes: The dependent variable is real, relative rent growth in year $t + 1$. Excess crowding is defined as the residual from the baseline rental density regression and is included after partialling out metropolitan-specific linear time trends using the Frisch–Waugh–Lovell theorem. All specifications include metropolitan and year fixed effects. Standard errors clustered at the metropolitan level are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table B5: Robustness: Metropolitan-Specific Linear Trends (Excluding 2020)

	(1) Real relative rent growth, $t+1$
Excess crowding	0.0319**
	(0.0160)
Metropolitan fixed effects	Yes
Year fixed effects	Yes
Metropolitan-specific linear trends	Yes
Pandemic year (2020) excluded	Yes
Observations	939
R^2 (within)	0.006

Notes: The dependent variable is real, relative rent growth in year $t+1$. Excess crowding is defined as the residual from the baseline rental density regression and is included after partialling out metropolitan-specific linear time trends using the Frisch–Waugh–Lovell theorem. The pandemic year (2020) is excluded from the sample. All specifications include metropolitan and year fixed effects. Standard errors clustered at the metropolitan level are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table B6: Dynamic Ordering Test: Excess Crowding and Subsequent Rent Growth

	(1) Real relative rent growth, $t+1$
Excess crowding (EC_t)	0.0408***
	(0.0134)
Lagged rent growth (ΔR_t)	0.3045***
	(0.0229)
Metropolitan fixed effects	Yes
Year fixed effects	Yes
Observations	1,900
R^2 (within)	0.101

Notes: The dependent variable is real, relative rent growth in year $t+1$. Excess crowding is measured in year t as the residual from the baseline rental density regression. Lagged rent growth is real, relative rent growth in year t. All specifications include metropolitan and year fixed effects. Standard errors clustered at the metropolitan level are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table B7: Dynamic Ordering Test: Rent Growth and Subsequent Excess Crowding

	(1)
	Excess crowding, $t+1$
Lagged rent growth (ΔR_t)	0.0020
	(0.0493)
Excess crowding (EC_t)	0.2109***
	(0.0374)
Metropolitan fixed effects	Yes
Year fixed effects	Yes
Observations	1,800
R^2 (within)	0.045

Notes: The dependent variable is excess crowding in year $t+1$. Lagged rent growth is real, relative rent growth in year t. Excess crowding is defined as the residual from the baseline rental density regression. All specifications include metropolitan and year fixed effects. Standard errors clustered at the metropolitan level are reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

References

Ahlfeldt GM, Pietrostefani E (2019) The economic effects of density: A synthesis. Journal of Regional Science 59(1):3–26

Albouy D, et al (2015) Driving to opportunity: Understanding the long-run decline in american mobility. AEA Papers and Proceedings 105(5):132–136

Calder-Wang S, Kim GH (2024) Coordinated vs efficient prices: The impact of algorithmic pricing on multifamily rental markets. Tech. rep., Cowles Foundation for Research in Economics, Yale University, URL https://cowles.yale.edu/sites/default/files/2024-05/Calder-Wang-main.pdf, working paper. Shows that RMS adoption drives occupancy toward a fixed 95–97% target, rendering occupancy a blunt demand metric

Davidoff T (2015) Supply constraints are not valid in high-cost areas. Critical Housing Analysis 2(1)

Duranton G, Puga D (2004) Micro-foundations of urban agglomeration economies. Handbook of Regional and Urban Economics 4:2063–2117

Fannie Mae (2023) Consumers feel renting is more affordable than homeownership. https://www.fanniemae.com/research-and-insights/surveys/national-housing-survey

Federal Reserve Bank of St. Louis (2024) Consumer price index for all urban consumers: Shelter vs all items less shelter. https://fred.stlouisfed.org/series/CUSR0000SA0L2

Feiveson L, et al (2024) Rent inflation and the role of demand. Board of Governors of the Federal Reserve System

Gabriel SA, Nothaft FE (2001) Rental housing markets, the incidence and duration of vacancy, and the natural vacancy rate. Journal of Urban Economics 49(1):121–149

Glaeser EL, et al (2001) Cities and skills. Journal of Labor Economics 19(2):316–342

Goodman AC (1992) Rental market adjustment and the natural vacancy rate. Journal of Urban Economics 31(1):1–16

Green RK, Malpezzi S, Mayo SK (2002) Measuring the elasticity of demand for housing services. Journal of Urban Economics 52(2):210–227

Liu S, Pan Q, Li Y (2018) Vertical cities: The price of tall buildings. Regional Science and Urban Economics 72:58–73

Malpezzi S, Green RK (1996) Rent control and rental housing quality: A note on the effects of new york city's 1971 law. Journal of Housing Economics 5(2):190–206

Molloy R (2022) Housing demand and supply: The need for more data and better policy. Brookings Papers on Economic Activity

Mott Z (2024) This region is finally seeing rent drop after a supply boom. Apartment List Research https://www.apartmentlist.com/research

Mueller GR, Laposa SP (1999) Real estate rental markets: What rental data can and cannot tell us. Journal of Real Estate Portfolio Management 5(2):135–144

Muth RF (1969) Cities and Housing. University of Chicago Press

Pennington K (2021) Does building new housing cause displacement? the supply and demand effects of construction in san francisco. Working Paper https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3902950

Pyhrr SA, Cooper JL, Wofford ML (1999) Real estate investment: Strategies, structures, decisions. Wiley

Ruggles S, Flood S, Sobek M, et al (2025) IPUMS USA: Version 16.0. https://doi.org/10.18128/D010.V16.0, URL https://doi.org/10.18128/D010.V16.0, dataset

Saiz A (2010) The geographic determinants of housing supply. Quarterly Journal of Economics 125(3):1253–1296

Sirmans C, Sirmans G, Benjamin J (1991) Determinants of apartment rent. Journal of Real Estate Research 6(3):357–379

The World Bank (2025) Population density (people per sq. km of land area). https://data.worldbank.org/indicator/EN.POP.DNST?most_recent_value_desc=true&locations=XD

Titman S, Wang K, Yildirmaz M (2024) City size and vertical development. Journal of Urban Economics 142:103519

U.S. Census Bureau (2022) Nearly half of renters are cost-burdened. https://www.census.gov/library/stories/2022/03/nearly-half-of-renter-households-are-cost-burdened.html

Wheaton WC (1991) Real estate "cycles": Some fundamentals. Real Estate Economics 19(2):209–230